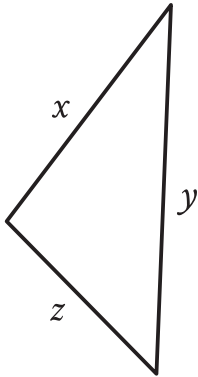


Question ID: 29e9b28c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Note: Figure not drawn to scale.

The triangle shown has a perimeter of **22** units. If $x = 9$ units and $y = 7$ units, what is the value of z , in units?

Answer

- A. **6**
- B. 7
- C. 9
- D. 16

Correct Answer: A

Rationale

Choice A is correct. The perimeter of a triangle is the sum of the lengths of its three sides. The triangle shown has side lengths x , y , and z . It's given that the triangle has a perimeter of **22** units. Therefore, $x + y + z = 22$. If $x = 9$ units and $y = 7$ units, the value of z , in units, can be found by substituting **9** for x and **7** for y in the equation $x + y + z = 22$, which yields $9 + 7 + z = 22$, or $16 + z = 22$. Subtracting **16** from both sides of this equation yields $z = 6$. Therefore, if $x = 9$ units and $y = 7$ units, the value of z , in units, is **6**.

Choice B is incorrect. This is the value of y , in units, not the value of z , in units.

Choice C is incorrect. This is the value of x , in units, not the value of z , in units.

Choice D is incorrect. This is the value of $x + y$, in units, not the value of z , in units.